

# Ganesh Gupta

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## EXPERIENCE

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### Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Engineered cloud-native executor routing pipeline nodes to **Docker** containers and serverless functions via **REST APIs**, implementing **OAuth**-based auth and **RBAC** for secure service access.
- Designed stateful recovery via per-node **Redis checkpointing**; interrupted workflows **resume from the last committed step** across **distributed execution**.
- Engineered pluggable **distributed executor** routing pipeline nodes to **Docker** or **AWS Lambda**, backed by **S3** artifact store eliminating redundant recomputation.
- Built **multi-agent system** via **OpenAI Agents SDK** + **MCP** tool-use handling **millions of daily transactions** at **99%+ uptime**.

### QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Designed and deployed a **Neo4j knowledge graph** modeling domain entities and relationships, powering **graph-aware retrieval** and reasoning in production.
- Cut **p99 inference latency 420ms to 140ms** across **50K+ daily requests** migrating synchronous serving to async **vLLM** on **GCP Cloud Run** with autoscaling.

### Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Designed and operated **25+ production Java microservices** processing **financial transactions** for **1M+ users**, maintaining **99.9% uptime** and **zero data loss** on AWS (EC2, RDS, SQS).
- Standardized **REST API** contracts across **25+ Java services** with consistent retry, timeout, and **OAuth**-aligned auth; reduced cross-service escalations **60%** within one quarter.
- Cut nightly financial data aggregation blocking downstream **settlement** from **8 minutes to 90 seconds** by redesigning SQL with **indexed views** and **batch partitioning**.

## PROJECTS

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### Distributed KV Store / VPC Control Plane Simulator

[github.com/ganeshhgupta/distributed-kv-store](https://github.com/ganeshhgupta/distributed-kv-store)

- Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion **VPC control plane** simulator modeling subnet routing and security groups.

### LangFetch

[tinyurl.com/LangFetch](https://tinyurl.com/LangFetch)

- LangGraph** agent decomposes multi-table questions into dependency-ordered **SQL subqueries** with intermediate result passing; deployable as **MCP-compatible** server.

### RootCause AI

[github.com/ganeshhgupta/rootcause-ai](https://github.com/ganeshhgupta/rootcause-ai)

- LangGraph** multi-agent system diagnosing **production incidents** via tool-use chains; routes alerts to specialized sub-agents and synthesizes **root cause reports**, reducing **MTTR**.

## EDUCATION

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### University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

### MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

## SKILLS

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**Languages:** Java, Python, Go, TypeScript, SQL, HTML/CSS

**Backend:** Identity and Access Control, Data Modeling, Large-scale Migrations, REST APIs

**Data and Infra:** PostgreSQL, Neo4j, Redis, Kafka, AWS, Kubernetes, Docker

**Practices:** Distributed Systems, System Design, Technical Leadership, Code Review

## CERTIFICATIONS

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AWS Certified Developer Associate | GCP Professional MLOps Engineer